PERSONAL PROTOTYPE

E-Rev Workbench

A small exploration of how spreadsheet-heavy environmental consulting review could be streamlined with transparent calculations first and AI-supported drafting second.

Environmental Consulting Review

Why I Built It

This prototype was inspired by a conversation about using AI to make routine environmental spreadsheet work faster. I wanted to make the idea concrete by building a local workflow that reads result tables, checks common issues, and produces review-ready outputs.

What It Does

- Reads CSV, TSV, TXT, and browser-supported Excel files.
- Maps common environmental fields such as location, date, matrix, analyte, result, units, depth, and coordinates.
- Flags QA review items and screens placeholder criteria.
- Builds exceedance tables, trend views, depth profiles, and map views.

Realistic Sample Snapshot

The included dataset is fictitious but shaped like a multi-matrix environmental review file, with groundwater, soil, soil gas, coordinates, qualifiers, reporting limits, depth intervals, and repeated sample dates.

992
RECORDS

893
DETECTIONS

157
EXCEEDANCES

110
QA FLAGS

Where AI Fits

The prototype does not rely on AI to invent conclusions or redo the math. E-Rev produces deterministic findings locally, then packages those findings into AI-ready prompts and draft-review checks. A production version could use a secure backend to generate memo language from verified findings.

Consulting Value

- Reduces repetitive spreadsheet sorting and manual summary work.
- Makes early QA checks more consistent.
- Keeps trend, depth-profile, and map outputs visible and auditable.
- Gives consultants faster starting points for interpretation and communication.

Production Build Path

- Verified federal, state, client, and project-specific criteria libraries.
- Secure AI drafting and review comments.
- Word/PDF exports aligned to company or client templates.
- Project storage, reviewer signoff, and versioned review packages.
- Expanded GIS and hydrogeology tools.

Limitations

This is a personal prototype, not a production or regulatory tool. It uses fictitious sample data and placeholder criteria. Trend labels and exceedance checks are first-pass review aids and should be confirmed with verified standards, project context, and professional judgment.

E-Rev Workbench is intended to demonstrate workflow thinking: auditable environmental review logic first, AI-supported narrative drafting second.